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Energy use pattern analyses of greenhouse vegetable production

M. Canakci, I. Akinci*

Department of Agricultural Machinery, Faculty of Agriculture, Akdeniz University, 07070 Antalya, Turkey

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Abstract

Greenhouse farming is a growing industry in many states. It is a very expensive way to produce greenhouse crops and there are many variables to consider before the farmer decides to take this route. A good location is essential for crop planning and growing. However, current studies related to energy use patterns and resources present in vegetable production are very limited. This research attempts to investigate the energy use patterns in greenhouse vegetable production, to determine the energy output–input ratio and their relationships. Antalya province, which has greenhouse area of about 13,337 ha (30.2%), is the center of greenhouse farming in Turkey. A questionnaire was distributed to 101 greenhouse farms from 11 villages in order to obtain the available data for vegetable production. Power requirement of the machines used in greenhouse operations were measured by using a computer based data acquisition system. Energy and economical variables (i.e. output–input ratio, specific energy, production cost, net return, etc.) were calculated by using the standard equations. As a result, the operational energy and energy source requirements of the greenhouse vegetable production were found between the ranges of 23,883.5–28,034.7 and 45,763.3–49,978.8 MJ/1000 m^2, respectively. The energy ratio of four major greenhouse vegetables—tomato, pepper, cucumber and eggplant—was 0.32, 0.19, 0.31, 0.23, respectively. The crop yields increased as a function of the total energy inputs with the best form being second-degree polynomial. The net return of the vegetable production was found in the 595.6–2775.3 \$/1000 m^2 ranges. Among the greenhouse vegetables, tomato cultivation resulted in being the most profitable.
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1. Introduction

Energy, being the capacity to do work, is at the heart of all human activities, especially those concerning the production of goods and services [1]. The demand for energy in agriculture has increased

* Corresponding author. Tel.: +90 242 310 24 64; fax: +90 242 227 45 64.
E-mail address: iakinci@akdeniz.edu.tr (I. Akinci).

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considerably with the introduction of high-yielding varieties and mechanised-crop production practices. Therefore, it is necessary to implement a switch from conventional to alternative sources of energy [2].

Several studies have been concentrated generally on worldwide production of field crops such as wheat, rice, soybean, cotton, maize, mustard, cluster bean, green gram, pearl millet, sugarcane, etc. in order to improve the energy output–input analyses and to investigate their relationships [3–11].

Hetz [12] studied on the utilization of energy of the fruit production in Chile in order to improve the efficiency of its use. Gezer et al. [13] studied on the energy input–output of the apricot production in the Malatya region of Turkey to optimise the energy sources. Onal and Tozan [14] analysed the energy requirements of the alternative production systems of the tomato processing. However, the studies related to the energy use patterns in vegetable production are very limited. More research is needed to document the energy use pattern and availability of resources in greenhouse vegetable production. Therefore, a study was undertaken with the following objectives:

- to investigate the energy use pattern in greenhouse vegetable production;
- to determine the energy output–input ratio and their relationships; and
- to analyse the economy of four major vegetable crops—tomato, pepper, cucumber and eggplant—produced in greenhouses.

2. Material and methods

Antalya province is one of the most important agricultural production areas and also, is the center of greenhouse vegetable production in Turkey. This region is located in the south–west of Turkey within 36°7′ and 37°26′ north latitude and 29°17′ and 32°44′ east longitude. The total area of Antalya province is about 2,059,101 ha which includes an agricultural area of 424,722 ha with a share of 20.62%. Different geographical and climatic characteristics increase the variety of crop patterns and greenhouse farms have a very important economical value in the province. Greenhouse farming is mostly concentrated on the south coast of the region. In this area, the coast line plains are surrounded by the Taurus Mountains in the north and by the Mediterranean Sea in the south. A typical Mediterranean climate is seen in the region with dry and hot summers and rainy and mild winters. Annual average temperature and total rainfall are 18.2 °C and 971.7 mm, respectively [15–17].

Agricultural operations, and the energy inputs and outputs values of greenhouse vegetable crops cultivated in the region, were determined by a questionnaire given to 101 farmers from 11 villages selected due to their regional properties. The Neyman method [18] was applied to determine farm numbers in Center, Kumluca, Kale and Kas districts between 2002 and 2003. First, the most common production systems for each crop were determined; then all inputs and outputs to and from the system were identified and quantified; later they were transformed into energy units. The energy equivalents of inputs and outputs are shown in Table 1.

The embodied energy and construction cost of a greenhouse was also estimated in respect of regional properties. It was said that typical greenhouse is a wide span type in the research region. The effective floor area of the greenhouse is 1008 m^2 (18×56 m). The height of the same is 2 m, but its center is raised to 5.5 m and the volume of greenhouse enclosure will be 3780 m^3. All surfaces of the greenhouses are

Table 1
Energy equivalences of inputs and outputs

Energy source	Units	MJ	Refs.
Human			
Man	h	1.96	[8,10,19]
Woman	h	1.57	[8,10,19]
N	kg	47.10	[20]
P_2O_5	kg	15.80	[20]
K_2O	kg	9.28	[20]
Diesel	l	47.8	[12]
Electric	kW h	10.59	[21]
Tractor	kg	93.61	[1]
Electric motor	kg	64.8	[8,10,19]
Agricultural machines	kg	62.7	[8,10,19]
Chemical	kg	120	[8,10]
Farmyard manure	kg	0.3	[8,10,22]
Plastics-general	kg	90	[23,24]
Wood	kg	18.9	[12]
Plant			
Tomato	Unit	0.28	Calculated
Pepper	Unit	0.28	Calculated
Cucumber		0.18	Calculated
Eggplant		0.28	Calculated
Output			
Tomato	kg	0.8	[10,22]
Pepper	kg	0.8	[10,22]
Cucumber	kg	0.8	[10,22]
Eggplant	kg	0.8	[10,22]

covered with glass of 3 mm thickness and its structural frame material is iron. The concrete wall of 0.25 m thickness was constructed on the perimeter of floor area as the foundation of the building. Table 2 gives the details of different materials used to construct the typical greenhouse [23–26]. In the research, initial investment values for construction of greenhouse were determined in respect of regional properties and questionnaire study. Cost break up of the greenhouse are shown in Table 3.

Table 2
Calculation of energy embodied for wide span greenhouse (18×56 m)

Item	Weight (kg)	Embodied energy (MJ/kg)	Total energy (MJ)
Iron	9000	27.73	249,570.0
Glass	10,865	12.70	137,985.5
Cement	1000	7.79	7790.0
Sand	42,000	0.08	3360.0
Screw	80	31.06	2484.8
Glass putty	1325	18.01	23,863.3
Paint	60	90.4	5424.0
Wire rod	90	12.5	1125.0
Total	–	–	431,602.6

Table 3
Cost break of different components of wide span greenhouse (18×56 m)

Item	Amount		Rate	Cost
	Quantity	Unit	$	$
Iron	9000	kg	0.46	4140.00
Glass	1440	m^2	6.69	9633.60
Cement	20	beg	2.93	58.60
Sand	42	t	2.48	104.16
Screw	80	kg	1.72	137.60
Glass putty	1325	kg	0.83	1099.75
Paint	60	kg	1.72	103.20
Wire rod	90	kg	1.38	124.20
Ground labour charge	–	–	137.93	137.93
Construction labour charge	–		793.1	793.10
Canopy labour charge	–	–	655.17	655.17
Total	–	–	–	16,987.32

In respect to regional properties, the greenhouse usage life was evaluated as 25 years. Under the circumstances, greenhouse embodied energy is 17,264.1 MJ per production area of 1008 m^2 for one year. This value is calculated at 17,127.1 MJ for 1000 m^2 greenhouse production area.

Energy is primarily used in agricultural operations for solarization, farmyard manure application, site preparation, transplanting, plant husbandry, spraying, irrigation, fertilizer application, heating, ventilation, harvesting, transportation and cleaning up.

Production energy of tractors and agricultural machines per unit area were calculated by using the following equation [5,13,14]

$$M_{\mathrm{pe}} = \frac{G \times M_{\mathrm{p}}}{T \times W} \tag{1}$$

where M_{pe} is the energy of the machine per unit area in MJ/1000 m^2, G is the mass of machine in kg, M_{p} is the production energy of the machine in MJ/kg, T is the economic life in h and W is the effective field capacity in 1000 m^2/h.

Diesel energy requirement was determined on the basis of fuel consumption and then, this value was converted into the energy units and expressed in MJ/1000 m^2. The following equation was used in the calculation of fuel consumption [5,27]

$$\mathrm{FC} = P_m \times R \times \mathrm{SFC} \tag{2}$$

where FC is the fuel consumption in l/h, P_m is the tractor power in kW, R is the loading ratio in decimal and SFC is the specific fuel consumption and taken as 0.30 l/kW h.

Power requirement and the loading ratio of the machines were measured by using a computer based data acquisition system. This system consisted of a dynamometer (50 kN), a torque meter (2000 N m), a data logger, a laptop PC and a connection frame. Electric consumption required for irrigation and spraying operations was estimated by an electricity counting meter.

Tomato, pepper, cucumber and eggplant plants were transplanted by hand and the farmers usually grow the plants themselves. Therefore, the energy equivalences for one plant was calculated by

combining the energy inputs, seed, farmyard manure, chemicals, nylon, human, and fertilizer, in the vegetable production. Energy output–input ratio, specific energy, energy productivity and energy intensiveness were calculated using the data recorded. In the stage of economic analyses, total cost included fixed and variable inputs were considered [26–30].

Table 4
Use of power source for greenhouse vegetable production

Operation	Tomato	Pepper	Cucumber	Eggplant
Solarization				
Man (h/1000 m^2)	41.6	41.6	41.6	41.6
Irrigation system (h/1000 m^2)	6.0	6.0	6.0	6.0
Electric (kW h/1000 m^2)	7.5	7.5	7.5	7.5
Nylon (kg/1000 m^2)	10.0	10.0	10.0	10.0
Farmyard manure application				
Man (h/1000 m^2)	28.3	28.1	27.7	28.7
Tractor (h/1000 m^2)	4.8	4.5	4.4	5.0
Diesel (l/1000 m^2)	10.3	9.7	9.4	10.8
Site preparation				
Man (h/1000 m^2)	48.9	48.9	48.9	48.9
Tractor (h/1000 m^2)	2.5	2.5	2.5	2.5
Diesel (l/1000 m^2)	6.3	6.3	6.3	6.3
Transplanting				
Man (h/1000 m^2)	28.6	20.6	32.0	22.9
Plant husbandry				
Man (h/1000 m^2)	1064.3	414.5	273.4	1337.6
String (kg/1000 m^2)	16.8	30.2	18.8	33.6
Spraying				
Man (h/1000 m^2)	26.0	20.7	38.4	25.0
Sprayer (h/1000 m^2)	26.0	20.7	38.4	25.0
Electric (kW h/1000 m^2)	25.0	19.9	36.9	24.0
Irrigation and fertilization				
Man (h/1000 m^2)	16.0	16.0	16.0	16.0
Irrigation system (h/1000 m^2)	90.1	72.4	92.7	83.6
Electric (kW h/1000 m^2)	112.6	90.5	115.9	104.5
Heating				
Man (h/1000 m^2)	81.2	80.0	78.9	81.8
Machine (h/1000 m^2)	81.2	80.0	78.9	81.8
Wood (kg/1000 m^2)	770.0	759.0	748.0	775.5
Ventilation				
Man (h/1000 m^2)	16.7	16.7	16.7	16.7
Harvesting				
Man (h/1000 m^2)	304.0	302.4	348.0	716.1
Transportation				
Man (h/1000 m^2)	16.8	20.2	50.5	27.8
Tractor (h/1000 m^2)	16.8	20.2	50.5	27.8
Diesel (l/1000 m^2)	24.0	25.7	62.3	35.3
Cleaning up				
Man (h/1000 m^2)	33.6	38.8	20.8	53.5

The use of power sources for the vegetable production in greenhouses are given in Table 4. The agricultural implements used in the greenhouse operations were driven by a tractor or an electric motor. Both the harvesting and cleaning up operations were done by hand only. No animal power was used for all greenhouse applications during the production process.

For all vegetable crops, solarization was applied for soil sterilization between July and August. This operation involves covering the surface of the soil with nylon material and setting up of a drip irrigation system. Then the soil was irrigated approximately 6 times during the soil sterilization. Finally, both the nylon material and the drip irrigation system were recovered from the soil surface before the site preparation.

The application of farmyard manure is generally applied on the vegetable crop site annually before site preparation. For the manure application the manpower, tractor-machinery and diesel fuel requirements were found at about 28, 4.7 h/1000 m^2 and 10 l/1000 m^2, respectively.

For the site preparation, the agricultural implements i.e. mouldboard plough, chisel and rotary tiller were used. Before transplanting, the drip irrigation system was installed by hand. Similar operations were applied in the site preparations of the other vegetable crops.

The manpower needed for transplanting operation varied from 20.6 to 32.0 h/1000 m^2 depending upon the number of plants per thousand square meters.

During the plant husbandry process manpower is used in the hoeing, weeding, pruning, wiring, training and pollination operations. The values of manpower consumed in the eggplant and tomato production were higher than the other crops. These values were 1337.6 and 1064.3 h/1000 m^2, respectively. This result was attributed to the high manpower usage for the pruning and pollination operations.

A two-wheel sprayer, which is driven by an electric motor and has a chemical tank of 200 l capacity, was used for spraying the vegetable crops. For irrigation of the vegetable crops, the drip irrigation systems were most commonly used in the greenhouses. Also the fertilizer applications were combined with those applications.

The traditional wood stoves manufactured by sheet iron were used for heating the greenhouses during winter. This operation is considered essential to avoid crop destruction when the temperature drops below 0 °C in the greenhouses. For this reason, the heating values were very close to each other. In the farmer greenhouses, there are no cooling systems or efficient ventilation systems. Natural ventilation systems were used in the greenhouses and their windows were opened or closed by hand to control the ventilation.

Manpower is used to harvest the vegetable crops and the number of operations increased during the harvesting period of May and June. The average distance from the greenhouses to the consumer markets is approximately 4000 m. The transportation time of the cucumber plant was higher than the other crops because of high crop yield with more harvesting number. At the end of the growing season plant stalks and leaves were cleaned up and carried out from the greenhouses, these operations were completed by hand.

3. Results and discussion

3.1. Operational energy

The operational energy requirements for the tomato, pepper, cucumber and eggplant production were 24,966.5, 23,883.5, 26,900.7 and 28,034.7 MJ/1000 m^2, respectively (Table 5). The wood material with a high-energy value was used for heating the greenhouses. The heating energy varied from 14,481.9 to

Table 5
Energy use pattern for greenhouse vegetable production

	Tomato	Pepper	Cucumber	Eggplant
A. Operations (MJ/1000 m^2)				
Solarization	836.0 (3.3)[a]	836.0 (3.5)	836.0 (3.1)	836.0 (3.0)
Farmyard manure application	761.6 (3.1)	729.1 (3.1)	717.6 (2.7)	784.8 (2.8)
Site preparation	467.9 (1.9)	467.9 (2.0)	467.9 (1.7)	467.9 (1.7)
Transplanting	50.5 (0.2)	36.4 (0.2)	56.5 (0.2)	40.4 (0.1)
Plant husbandry	2634.5 (10.6)	2090.6 (8.8)	1328.6 (4.9)	3872.8 (13.8)
Spraying	384.6 (1.5)	306.2 (1.3)	568.0 (2.1)	369.8 (1.3)
Irrigation and fertilization	2309.5 (9.3)	1863.8 (7.8)	2375.0 (8.8)	2145.8 (7.7)
Heating	14,902.2 (59.7)	14,692.1 (61.5)	14,481.9 (53.8)	15,007.3 (53.5)
Ventilation	29.5 (0.1)	29.5 (0.1)	29.5 (0.1)	29.5 (0.1)
Harvesting	595.8 (2.4)	592.7 (2.5)	682.1 (2.5)	1403.6 (5.0)
Transportation	1928.5 (7.7)	2163.2 (9.1)	5316.8 (19.8)	2971.9 (10.6)
Cleaning up	65.9 (0.3)	76.0 (0.3)	40.8 (0.2)	104.9 (0.4)
Total	24,966.5 (100.0)	23,883.5 (100.0)	26,900.7 (100.0)	28,034.7 (100.0)
B. Sources (MJ/1000 m^2)				
Human	3127.4 (6.6)	1967.0 (4.3)	1883.2 (3.8)	4467.8 (8.9)
Plant	700.0 (1.5)	504.0 (1.1)	504.0 (1.0)	560.0 (1.1)
Machine	1300.6 (2.8)	1433.4 (3.1)	2826.8 (5.7)	1782.2 (3.6)
Plastics	2503.6 (5.3)	2897.5 (6.3)	2624.3 (5.3)	3182.8 (6.4)
Diesel	1945.5 (4.1)	1992.2 (4.4)	3732.2 (7.5)	2504.7 (5.0)
Electric	1536.5 (3.3)	1248.3 (2.7)	1696.9 (3.4)	1440.2 (2.9)
Fertilizer	2129.4 (4.5)	2253.7 (4.9)	2551.6 (5.2)	2175.0 (4.4)
Farmyard manure	1008.0 (2.1)	975.0 (2.1)	963.0 (1.9)	1050.0 (2.1)
Chemical	1164.0 (2.5)	1020.0 (2.2)	1392.0 (2.8)	1032.0 (2.1)
Wood	14,553.0 (30.9)	14,345.1 (31.3)	14,137.2 (28.6)	14,657.0 (29.3)
Greenhouse structure	17,127.1 (36.4)	17,127.1 (37.4)	17,127.1 (34.6)	17,127.1 (34.3)
Total	47,095.1 (100.0)	45,763.3 (100.0)	49,438.3 (100.0)	49,978.8 (100.0)
C. Others				
Crop yield (kg/1000 m^2)	18,895	10,598	19,201	14,376
Energy ratio (decimal)	0.32	0.19	0.31	0.23
Specific energy (MJ/kg)	2.492	4.318	2.575	3.477
Energy productivity (kg/MJ)	0.40	0.23	0.39	0.29

[a] Figure in parentheses indicates a percentage of the energy inputs.

15,007.3 MJ/1000 m^2 (60% of the total energy in operations) due to the different geographical and climatic conditions.

The maximum use of the transportation energy was 5316.8 MJ/1000 m^2 in the cucumber production followed by the eggplant vegetable (2971.9 MJ/1000 m^2) due to more number of harvesting and transportation numbers. The irrigation and fertilization energy consumed by the tomato, pepper, cucumber and eggplant productions were 2309.5, 1863.8, 2375.0 and 2145.8 MJ/1000 m^2, respectively. The energy requirements for the tomato and cucumber productions were slightly greater than the pepper and eggplant cultivations.

The plant husbandry energy of the eggplant and tomato productions was significantly greater than the pepper and cucumber energies because of high manpower use (Table 5). The energy in farmyard manure application was in 717.6–784.8 MJ/1000 m^2 range with a share of approximately 3%. These energy

values were found very close to each other because of the equal farmyard manure application. The spraying energy varied from 306.2 to 568.0 MJ/1000 m^2 with a share of 2.2%.

It was observed that the harvesting energy was approximately 2.5% of total operational energy except for the eggplant harvesting. The low harvesting capacity increased the energy ratio with a share of 5.0%. The solarization, site preparation and ventilation energies were estimated at 836.0, 467.9 and 29.5 MJ/1000 m^2, respectively. The energy use in transplanting, ventilation and the cleaning up were lower as compared to the other energy inputs.

3.2. Energy use for various sources

The total energy requirements from different sources for producing the tomato, pepper, cucumber and eggplant crops were 47,095.1, 45,763.3, 49,438.3 and 49,978.8 MJ/1000 m^2, respectively (Table 5). Among the different energy sources, the greenhouse structure and wood energies were the highest with a mean of 17,127.1 and 14,423 MJ/1000 m^2, respectively, and their ratio was approximately 70%. These high values of both the greenhouse structures and wood energies can affect the ratio of other energy sources in the total inputs. In the research region, the greenhouses, which were covered with the glass material, had a high-energy value. Therefore, this material significantly increased the structure energy of the greenhouses. This result must be considered in the energy efficiency of greenhouse farming.

Because of the high manpower used in the plant husbandry and harvesting operations, the human energy of the eggplant production had the biggest share of 8.9% of the total. The tomato, pepper and cucumber energies were found with a share of 6.6, 4.3 and 3.8%, respectively. The fertilizer energy was in the 2129.4–2551.6 MJ/1000 m^2 ranges. These values were found very close to each other in respect of using the same amount of the fertilizing.

The diesel energy of the cucumber production (3732.2 MJ/1000 m^2) was greater than the other crops. This result could be attributed to the high crop yield and more number of transportation operations. The machine energy used in the vegetable production varied from 1300.6 to 2826.8 MJ/1000 m^2. The numbers of the transportation and spraying operations in the cucumber producing increased the machine energy with a share of 5.7%. The electric energy consumed in the four vegetable productions varied from 1248.3 to 1696.9 MJ/1000 m^2. The highest value was obtained by the cucumber production with a share of 3.4% caused by the high-powered electric motor used for spraying and irrigation. Plastic energy was found in the 2503.6–3182.8 MJ/1000 m^2 ranges. With the same types irrigation system in all greenhouses, these values varied fractionally.

The plant energy consumed in the four vegetable transplanting varied between 504.0 and 700.0 MJ/1000 m^2. In general review, this type of energy source accounted for a low amount of the total inputs. This result could be attributed to the plant number per unit area.

Farmyard manure is used for the greenhouse vegetable production annually. The average farmyard manure energy was found at about 1000 MJ/1000 m^2. The chemical energy varied from 1020.0 to 1392.0 MJ/1000 m^2. Its ratio in the total inputs was about 2.5%. The chemical energy ratio in the cucumber production was higher than the other crops.

In these circumstances, efficient use of these energy sources helped to achieve increased production and productivity and contributed to economical and therefore higher profitability of greenhouse agriculture.

3.3. Other energy variables

The vegetable yields of the tomato, pepper, cucumber and eggplant crops were determined at 18,895, 10,598, 19,201, and 14,376 kg/1000 m^2, respectively. It was found that the energy ratio had the highest value in the tomato production followed by cucumber, eggplant and pepper vegetables. These values were calculated at 0.32, 0.31, 0.23 and 0.19, respectively. The specific energy requirement was the highest in the pepper production (4.318 MJ/kg) followed by the eggplant (3.477 MJ/kg), cucumber (2.575 MJ/kg) and tomato (2.492 MJ/kg) cultivations. The energy productivity in the greenhouse vegetable production varied from 0.23 to 0.40 kg/MJ. The highest value being for the tomato cultivation. The output energy values of crops are different due to crop yield. Therefore, although there were no significant differences among the total energy input values, the energy ratio values of the crops were pretty different. Also, the differences affect the specific energy and energy productivity values. The differences are clearer between tomato, cucumber values and pepper value.

The average energy ratio of potato and vegetables and tomato grown in open field were found at 0.58 and 0.83–1.62, respectively [3,14]. In comparison with the previous studies, the energy ratio of the greenhouse vegetable production was found to have lower values varied from 0.19 to 0.32. This result could be attributed to the high structure and heating energies used in greenhouse farming. However, these results could not be compared due to lack of data relating to energy operations and energy sources in greenhouse vegetable production. From this prospective, this study will be greatly contributed to the future research into energy analyses of greenhouse vegetable production.

4. Energy input and crop yield relations

For the tomato, pepper, cucumber and eggplant production, the plots of predicted and observed values of the total energy input versus the crop yield are presented in (Figs. 1–4). The regression coefficients in relationship between the total inputs and the crop yields and the corresponding R^2 values are given in the Figures.

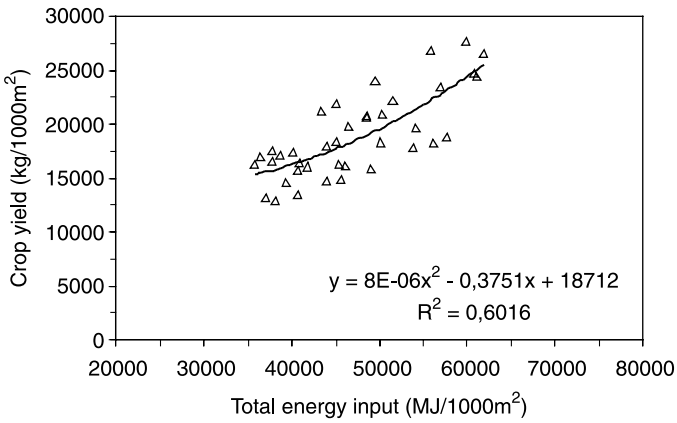

Fig. 1. Crop yield versus total energy input for tomato crops.

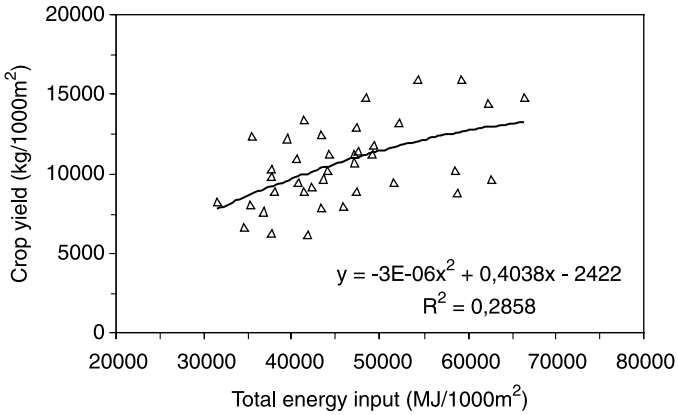

Fig. 2. Crop yield versus total energy input for pepper crops.

The best relationship between the total energy inputs and the crop yields were expressed as the best form of second degree polynomial regression. The crop yields in greenhouses increased as a function of the energy inputs (Figs. 1–4). The coefficient of determination (R^2) between yield and total energy input for tomato, pepper, cucumber and eggplant were 61.0, 28.1, 50.3 and 37.1%, respectively. The variation in data for the same crops can be explained on the basis of energy used through heating. There are no significant differences except wood among the input values for the same crops. Some farmers have to use more wood to prevent freeze. Similar trends were reported by Singh et al. [9,11] for some field crops i.e. cluster bean, wheat, cotton, maize, etc. Singh et al. [7] for cotton crops and Singh et al. [31] for rice crops.

5. Economic analyses

The total variable costs for tomato, pepper, cucumber and eggplant production were 2956.5, 2551.8, 3314.1 and 3820.6 \$/1000 m^2, respectively. The worker and fertilizer costs were greater than the other

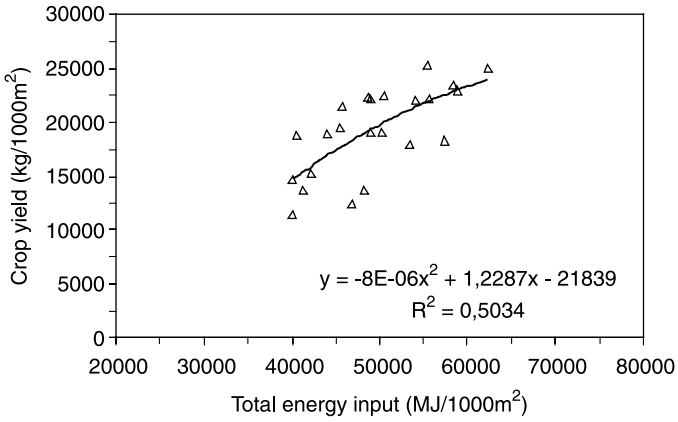

Fig. 3. Crop yield versus total energy input for cucumber crops.

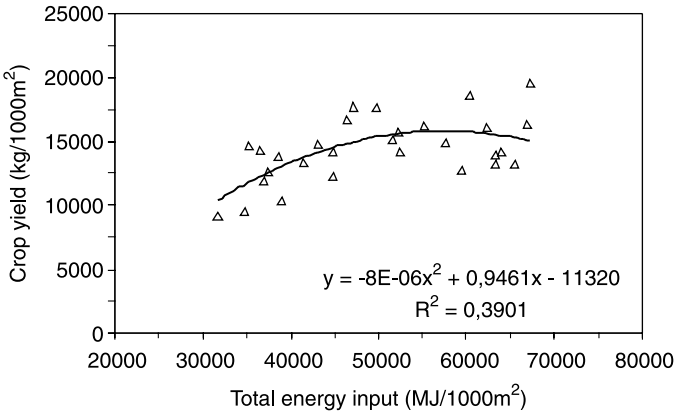

Fig. 4. Crop yield versus total energy input for eggplant crops.

components because of high manpower and fertilizer use in the greenhouse vegetable production. The worker cost for eggplant was the highest (1643.9 \$/1000 m^2) and tomato (1161.6 \$/1000 m^2) followed by pepper (706.1 \$/1000 m^2) and cucumber (646.8 \$/1000 m^2). The fertilizer costs for eggplant tomato, pepper, cucumber and eggplant production were 706.7, 663.3, 1133.3 and 906.7 \$/1000 m^2, respectively. These component followed by chemicals, plants, operating interest charges, farmyard manure and wood. The values other item were lower than 50 \$/1000 m^2 (Table 6). In the regional greenhouse, heating system is used to prevent only from freeze in the winter time. There are no modern heating, cooling, and ventilation systems, etc. in the farmer greenhouses. For this reason, diesel, electric, repair and maintenance costs were remained on low values. During the greenhouse vegetable production, glasses (covering material) can be broken by accident. The greenhouse repair cost contained replacement of the broken glasses and its labour cost. In the questionnaire study, this value were determined at 48.3 \$/1000 m^2 per year. Besides, if it is necessary the farmers clean glasses and replace glass putty within 10–15 years.

For the production of tomato, pepper, cucumber and eggplant crops, the total fixed cost were calculated at 1894.9, 1971.9, 1915.2 and 1944.1 \$/1000 m^2. The total fixed costs included greenhouse building and the other items. The depreciation, interest and land rent costs of greenhouse building were 495.2, 611.5 and 275.9 \$/1000 m^2, respectively. The other; depreciation, interest and housing defined machinery fixed costs.

As shown in Table 6, the eggplant production was the most investment intensive crop (5764.7 \$/1000 m^2) followed by the cucumber (5229.3 \$/1000 m^2), tomato (4851.4 \$/1000 m^2) and pepper (4423.7 \$/1000 m^2). The total and net returns of the tomato was higher than the other crops with a value of 7626.7 and 2775.3 \$/1000 m^2, respectively. The benefit/cost ratio for the tomato, pepper, cucumber and eggplant production were calculated at 1.57, 1.15, 1.29 and 1.10, respectively. In these circumstances, it was seen that the tomato production in greenhouses was found to be more profitable in point of both energy and economical values.

In the cultivation of some field crops i.e. wheat, chickpea, soybean, rice, pigeon pea etc. the net return values varied from 2.39 to 27.0 \$/1000 m^2 [4,8]. In comparison with these results, the net return values of vegetable production in greenhouses (595.6–2775.3 \$/1000 m^2) were found significantly greater than

Table 6
Economic analyses of greenhouse vegetable production

Cost items	Tomato	Pepper	Cucumber	Eggplant
Variable costs ($/1000) m^2				
Plant	224.1	335.2	386.2	220.7
Farmyard manure	92.7	89.7	88.6	96.6
Fertilizer	706.7	663.3	1133.3	906.7
Chemicals	366.7	336.7	540.0	433.3
Diesel	33.7	34.5	64.6	43.4
Lubricant	5.1	5.2	9.7	6.5
Wood	53.1	52.3	51.6	53.5
Worker	1161.6	706.1	646.8	1643.9
Operator	29.8	34.1	75.7	45.2
Electric	10.0	8.1	11.1	9.4
Plastic cover	10.3	10.3	10.3	10.3
String	41.7	75.0	46.7	83.4
Repair and maintenance	23.9	24.6	34.4	27.1
Greenhouse repair	48.3	48.3	48.3	48.3
Operating interest charges	148.8	128.4	166.8	192.3
Total	2956.5	2551.8	3314.1	3820.6
Fixed costs ($/1000 m^2)				
Greenhouse building				
Depreciation	495.2	495.2	495.2	495.2
Interest	611.5	611.5	611.5	611.5
Land rent	275.9	275.9	275.9	275.9
Others				
Depreciation	76.0	76.0	76.0	76.0
Interest	201.1	201.1	201.1	201.1
Housing	66.7	66.7	66.7	66.7
General overhead costs	168.5	145.5	188.8	217.7
Total	1894.9	1871.9	1915.2	1944.1
Total cost of production ($)	4851.4	4423.7	5229.3	5764.7
Total return ($)	7626.7	5074.2	6749.4	6360.3
Net return ($/1000 m^2)	2775.3	650.5	1520.1	595.6
Benefit/cost ratio	1.57	1.15	1.29	1.10

the field crops. These results must be considered in crop planning and energy requirements in agricultural farms to determine the economic viability of production.

6. Conclusions

(1) In greenhouse vegetable production, the solarization, farmyard manure application, site preparation, transplanting, plant husbandry, spraying, irrigation and fertilization, heating, ventilation, harvesting, transportation and cleaning up operations were applied by using the tractor, electric motor and manpower. No animal power was used for all greenhouse applications during the production process.

(2) The operational energy and energy source requirements of the vegetable production were found in the ranges of 23,883.5–28,034.7 and 45,763.3–49,978.8 MJ/1000 m^2, respectively.
(3) The energy ratio of four major greenhouse vegetables used in this study i.e.: tomato, pepper, cucumber and eggplant, were calculated at 0.32, 0.19, 0.31, 0.23, respectively.
(4) The best relationship between the total energy inputs and crop yields were expressed as the form of second degree polynomial regression. Therefore, the crop yields increased as a function of the total energy inputs.
(5) The total cost and net return of the vegetable production was found in the ranges 4423.7–5764.7 and 595.6–2775.3 \$/1000 m^2, respectively. In all vegetables, the tomato production resulted in being most profitable.

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